

B. Tech Degree VII Semester Examination November 2010

EB/EC/EI 705 (D) MECHATRONICS (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART – A (Answer ALL questions)

(8 x 5 = 40)

- I. (a) Write short notes on Elements of Mechatronic Systems.
(b) What is Hall effect? Explain Hall effect sensors.
(c) With diagram explain Ratchet and Pawl Mechanism.
(d) What is the need for bearing? Explain different types of ball bearings.
(e) Describe the basic building blocks of fluid flow system.
(f) Describe the output input relationship of Rotational translational systems.
(g) With necessary diagrams and circuits explain proportional plus derivative control.
(h) Explain the working of Robotic Systems.

PART – B

(4 x 15 = 60)

- II. (a) Explain the specifications of sensors and transducers. (10)
(b) Explain the working of proximity sensors. (5)
OR
III. (a) With block diagram explain Data Acquisition Systems. (10)
(b) What is signal conditioning? Explain error compensation. (5)

- IV. (a) With neat sketch explain the working of process control valves. (10)
(b) Write short notes on Kinematic chains. (5)

OR

- V. Explain the working of various Electrical Actuation Systems. (15)

- VI. (a) Write short notes on Dynamic response of systems. (5)
(b) Explain the output input relationship of Hydraulic Mechanical Systems. (10)

OR

- VII. What are MEMS? Explain various manufacturing process of MEMS. List the advantages and applications of MEMS. (15)

- VIII. With the help of block diagram explain the structure and I/o processing of Programmable Logic Controllers. (15)

OR

- IX. Write short notes on the following.
(i) AGV
(ii) FMC
(iii) FMS (15)